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[2]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.svm import SVC
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
import seaborn as sns
import matplotlib.pyplot as plt

[8]: df = pd.read_csv("netflix_titles.csv")

[9]: df["director"] = df["director"].fillna("Unknown")
df["cast"] = df["cast"].fillna("Unknown")
df["country"] = df["country"].fillna("Unknown")
df["rating"] = df["rating"].fillna(df["rating"].mode()[0])
df["duration"] = df["duration"].fillna(df["duration"].mode()[0])

[10]: df = df[["type", "release_year", "rating", "duration"]]

[11]: le = LabelEncoder()

df["type"] = le.fit_transform(df["type"]) # Target
df["rating"] = le.fit_transform(df["rating"])
df["duration"] = le.fit_transform(df["duration"])

[12]: X = df.drop("type", axis=1)
y = df["type"]

[13]: scaler = StandardScaler()
X = scaler.fit_transform(X)

# Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
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[14]: svm = SVC(kernel="linear")
svm.fit(X_train, y_train)

y_pred_svm = svm.predict(X_test)

print("SVM Accuracy :", accuracy_score(y_test, y_pred_svm))
print("SVM Confusion Matrix")
print(confusion_matrix(y_test, y_pred_svm))

# ----- Random Forest -----
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)

print("\nRandom Forest Accuracy :", accuracy_score(y_test, y_pred_rf))
print("Random Forest Confusion Matrix")
print(confusion_matrix(y_test, y_pred_rf))

SVM Accuracy : 0.746311010215664
SVM Confusion Matrix
[[1006 208]
 [ 239 309]]

Random Forest Accuracy : 0.9772985244040863
Random Forest Confusion Matrix
[[1198  16]
 [  24 524]]
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[15]: plt.figure(figsize=(10,4))

plt.subplot(1,2,1)
sns.heatmap(confusion_matrix(y_test, y_pred_svm),
            annot=True,
            fmt="d",
            cmap="Blues")
plt.title("SVM")

plt.subplot(1,2,2)
sns.heatmap(confusion_matrix(y_test, y_pred_rf),
            annot=True,
            fmt="d",
            cmap="Greens")
plt.title("Random Forest")

plt.tight_layout()
plt.show()
```

